

Unit1

Unit 1

1. Explain the major hardware components of a computer with their functions.
2. Describe the main functions of an operating system with examples.
3. Discuss the different types of software with suitable examples.
4. Explain the importance of networking. Compare LAN and WAN.
5. With a neat diagram, explain the architecture of a computer system.
6. Discuss the role of peripherals in computer usage and give suitable examples.
7. Explain different types of computer storage devices with examples.
8. What is the internet? Mention one of its uses.
9. Differentiate between RAM and ROM.
10. What is a motherboard? Mention its importance.
11. Explain the difference between system software and application software.

★ Multiple Choice Questions (MCQs)

1. The “brain” of the computer is:
 - a) RAM
 - b) CPU**
 - c) Hard disk
 - d) Monitor
2. Which of the following is a volatile memory?
 - a) ROM
 - b) Hard Disk
 - c) RAM**
 - d) Pen drive
3. Which of the following is NOT an input device?
 - a) Keyboard
 - b) Printer**
 - c) Mouse
 - d) Scanner
4. Operating system is:
 - a) Hardware
 - b) Application software
 - c) System software**
 - d) None of these

5. LAN stands for:
- a) Local Access Network
 - b) Local Area Network**
 - c) Large Area Network
 - d) Line Access Network
6. Which device is used to permanently store data?
- a) RAM
 - b) Cache
 - c) Hard disk**
 - d) Register
7. WAN covers:
- a) Small office
 - b) Single building
 - c) City
 - d) Large geographical area**
8. Which component connects all hardware parts together?
- a) CPU
 - b) Motherboard**
 - c) RAM
 - d) Cache
9. Example of application software is:
- a) MS Word**
 - b) Windows 10
 - c) Linux
 - d) DOS
10. The device that displays the output visually is:
- a) Keyboard
 - b) Mouse
 - c) Monitor**
 - d) Speaker

Unit-2

1. Explain the systematic steps of troubleshooting with an example.
2. Write common computer problems and their possible causes with solutions.
3. What is BIOS/UEFI? Explain their functions and differences.
4. Define diagnostic tools. List types with suitable examples.
5. Explain the role of system information utilities in troubleshooting.
6. Describe common hardware issues like overheating, noise, and loose connections.
7. Explain safe handling practices of computer components. Why is ESD protection important?
8. Write short notes on:
 - a) Overheating issues
 - b) CMOS error
 - c) Frozen screen
9. Explain how Task Manager, CHKDSK, and Windows Memory Diagnostic are used for troubleshooting.

★ Multiple Choice Questions (MCQs)

- 1) The first step in troubleshooting is:
 - a) Solution Testing
 - b) Information Gathering
 - c) **Problem Identification**
 - d) Implementation
- 2) Which of the following is NOT a common computer problem?
 - a) Blank screen
 - b) Overheating
 - c) Blue Screen of Death
 - d) **Data Encryption**
- 3) Which tool is used to check hard disk errors in Windows?
 - a) Disk Cleanup
 - b) Task Manager
 - c) **CHKDSK**
 - d) Device Manager
- 4) BIOS stands for:
 - a) **Basic Input/Output System**
 - b) Binary Integrated Operating Setup

- c) Boot Input/Output System
- d) Basic Internal Operating Software

5) UEFI is:

- a) Older firmware used in old PCs
- b) A modern replacement for BIOS**
- c) A type of operating system
- d) A computer virus detection tool

6) Which diagnostic tool provides real-time data about CPU, memory, and disk usage?

- a) Event Viewer
- b) Task Manager**
- c) BIOS
- d) Device Manager

7) Strange clicking noises inside the CPU usually indicate:

- a) Overheating RAM
- b) Hard drive failure**
- c) Loose power cable
- d) Virus infection

8) To prevent Electrostatic Discharge (ESD) while handling components, one should:

- a) Wear rubber shoes
- b) Use an ESD wrist strap**
- c) Work on a carpet
- d) Keep the computer turned on

9) POST is performed by:

- a) Operating System
- b) BIOS/UEFI**
- c) Hard Disk
- d) Application Software

Unit:-3

1. What is File Management? Describe its key functions, importance, and tools used in different operating systems.
2. Discuss different types of system updates (security, performance, feature, driver) and explain their importance with examples.
3. Explain the role of OS security settings. Describe tools like antivirus, firewall, user accounts, and authentication in maintaining system security.
4. Write about common software issues such as application crashes, outdated drivers, insufficient resources, and malware. Mention their causes and troubleshooting methods.
5. Describe safe practices for software installation and uninstallation. Also explain system recovery tools (System Restore, Safe Mode, Reset This PC, Backup & Restore) with examples.

1. The main function of an operating system is:

- a) Weather Forecasting
- b) File and Process Management**
- c) Designing Graphics
- d) Video Editing

2. Which tool is used in Windows for file management?

- a) Task Manager
- b) File Explorer**
- c) Device Manager
- d) Disk Cleanup

3. Updates released to fix security vulnerabilities are called:

- a) Performance Updates
- b) Security Updates**
- c) Driver Updates
- d) Feature Updates

4. A tool that blocks suspicious or harmful network traffic is:

- a) Calculator
- b) Firewall**
- c) Task Manager
- d) File Explorer

5. Which of the following is a common symptom of an application crash?

- a) Faster performance

b) "Program not responding" message

- c) Extra storage space
- d) Automatic update installation

6. Which Windows tool is used to update or reinstall drivers?

- a) Disk Cleanup
- b) Device Manager**
- c) File Explorer
- d) Control Panel → Fonts

7. Downloading software from untrusted sources can cause:

- a) Faster updates
- b) Virus/Malware infection**
- c) More free space
- d) Increased performance

8. Which of the following is a system recovery tool in Windows?

- a) MS Word
- b) System Restore**
- c) Notepad
- d) Disk Defragmenter

9. Storing backup on Google Drive is an example of:

- a) External Backup
- b) Cloud Backup**
- c) Incremental Backup
- d) Full Format

10. Safe Mode in Windows loads:

- a) All drivers and applications
- b) Only essential drivers and services**
- c) No drivers at all
- d) Antivirus software only

Unit:-4

1. Explain the importance of routine maintenance for a computer system. Mention any two tools used and their benefits.
2. What is malware? Describe any two types of malware with suitable examples.
3. List any four basic security principles that help protect a computer from threats.
4. Differentiate between antivirus and anti-malware software. Why are both important for system security?
5. Mention three safe browsing practices and two data protection strategies that users should follow.

1. What is the main purpose of Disk Cleanup?

- A. To install updates
- B. To remove temporary and unnecessary files**
- C. To scan for viruses
- D. To rearrange files on the hard drive

2. Which of the following best describes disk defragmentation?

- A. Backing up data to a USB
- B. Deleting unwanted files
- C. Organizing scattered file pieces on the hard drive**
- D. Encrypting files for security

3. Why are software updates important for your computer?

- A. They slow down the system
- B. They increase file size
- C. They fix bugs and improve security**
- D. They remove antivirus

4. Which of the following is NOT a type of malware?

- A. Virus
- B. Trojan Horse
- C. Backup**
- D. Ransomware

5. A program that looks safe but steals your data is known as:

- A. Worm
- B. Spyware
- C. Trojan Horse**
- D. Ransomware

6. Which of the following is a good example of a strong password?

- A. ganpat123
- B. 12345678
- C. Ganpat@2025#IT**
- D. password

7. What is the function of anti-malware software?

- A. Only protects USB devices
- B. Only removes viruses
- C. Stops spyware, ransomware, and trojans**
- D. Defragments the hard disk

8. Which of the following is a safe browsing practice?

- A. Downloading cracked software
- B. Using public Wi-Fi for online banking
- C. Clicking unknown pop-ups
- D. Visiting only HTTPS websites**

Unit 5

1. Explain the different system optimization techniques used to improve computer performance without changing hardware.
2. Describe the importance of upgrading common computer components. How do RAM and SSD upgrades improve performance?
3. What are advanced troubleshooting resources? Explain how online forums, manufacturer support, and Windows troubleshooters help solve computer issues.
4. Write and explain safe practices for seeking technical help online. Why is it important to be careful while sharing personal or system information?
5. Differentiate between HDD and SSD. Which one is better for performance and why? Include at least three differences.

1. What does RAM stand for?
 - a) Read Access Memory
 - b) Random Access Memory
 - c) Ready Available Memory
 - d) Run Access ModuleAnswer: b) Random Access Memory

2. Which shortcut opens Task Manager directly?
 - a) Ctrl + Alt + Del
 - b) Ctrl + Shift + Esc
 - c) Alt + Tab
 - d) Ctrl + F5

Answer: b) Ctrl + Shift + Esc

3. Which tool is used to remove temporary files and free up disk space?
- a) Task Manager
 - b) Disk Cleanup
 - c) File Explorer
 - d) Device Manager

Answer: b) Disk Cleanup

4. What happens if too many startup programs are enabled?
- a) Computer boots faster
 - b) Boot time becomes slow
 - c) More storage space is created
 - d) System automatically updates

Answer: b) Boot time becomes slow

5. Which of the following is an advantage of SSD over HDD?
- a) Cheaper and slower
 - b) No moving parts, faster performance
 - c) Makes more noise
 - d) Consumes more power

Answer: b) No moving parts, faster performance

6. Which of the following is a safe source to download drivers or updates?
- a) Third-party website
 - b) Manufacturer's official website
 - c) Unknown forum
 - d) File-sharing sites

Answer: b) Manufacturer's official website

7. Which Windows feature can fix problems like no sound or network errors?
- a) Windows Defender
 - b) Task Manager
 - c) Windows Troubleshooter
 - d) Control Panel

Answer: c) Windows Troubleshooter

8. What should you NEVER share while seeking help online?
- a) Your operating system name
 - b) Passwords or OTPs
 - c) Computer brand
 - d) Model number

Answer: b) Passwords or OTPs